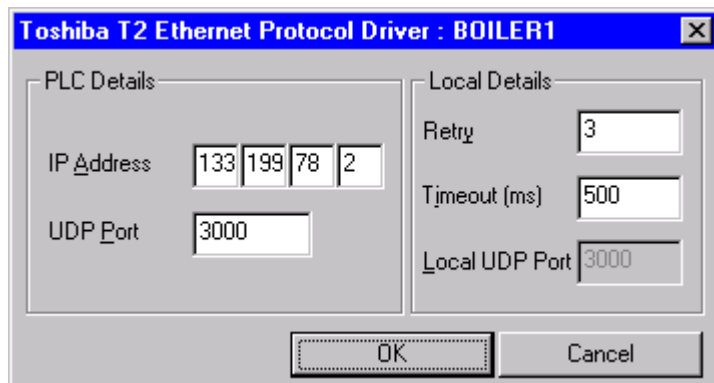


Toshiba T2 Ethernet protocol driver

Overview

The TOSHIBA Ethernet driver currently only supports the ascii-based protocol. This driver supports the PROSEC T2 PLC. Communication is UDP/IP. The ADROIT TOSHIBA driver will act as the master in a master-slave transaction system.



PLC Details

IP Address - This must be any available class B address on the network ie(OCTET1 must be > 127) setup in the PLC.

Port - The first available UDP port setup in the PLC

Local Details

Retry - This is the number of times the ADROIT TOSHIBA driver will attempt a particular scan job before failing.

Timeout - This is the maximum time the ADROIT TOSHIBA driver will wait for a response to its request message from the PLC.

Local UDP Port – **For indication only** of the Local UDP port used for this device. NOTE: This is not a fixed parameter it can change at startup depending on available ports.

Wiring

Standard Ethernet interconnection. (10BASE-T)

Supported Commands

Area (Register/ Device)	Format (Integer /Real)	Format (Boolean)	Description
XW	XW0	XW0.0 - XW0.15	External Input Register
YW	YW0	YW0.0 - YW0.15	External Output Register
ZW	ZW0	ZW0.0 - ZW0.15	Link Relay Register
RW	RW0	RW0.0 - RW0.15	Auxiliary Relay Register
D	D0	D0.0 - D0.15	Data Register
C	C0	C0.0 - C0.15	Counter Register
T	T0	T0.0 - T0.15	Timer Register

Message protocol

The checksum value is the ASCII code of the lower two digits of the sum of all the values of all the bytes in a message starting from the first byte until the "&" termination byte.

Telegram examples shown below. All values shown are in ASCII-HEX unless specified otherwise.

Read request:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Start Register / Device variable bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	DR	XW0	&	<<>>	)	CR

Read response:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Data variable bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	DR	<<>>	&	<<>>	)	CR

Write request:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Start Register / Device variable bytes	Data Delimiter	Number of data bytes	Data variable bytes	Checksum Delimite r	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	DW	XW0	,	1	<<>>	&	<<>>	)	CR

Write Acknowledgment:

Start Code	Format ID Code	Addr 2 bytes	Command 2 bytes	Status 4 bytes	Checksum Delimiter	CheckSum 2 bytes	End Code	Carriage Return
(	A	01	ST	0002	&	<<>>	)	CR

Revision information

Date:	Changes:
19-Sep- 2000	First release notes!